IGCSE 0607 Practice Questions — E8.3 — Combined Questions & Markscheme

Magnitude of a vector

Total Questions: 6 — **Total Marks:** 38 Marks

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1 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

2 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

3 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u}+\mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

4 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

5 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

6 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]

1 **(a)** $\sqrt{(-8)^2+15^2}$ or $\sqrt{64+225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p}+2\mathbf{q}=\begin{pmatrix}-8\\15\end{pmatrix}+\begin{pmatrix}4\\-8\end{pmatrix}=\begin{pmatrix}-4\\7\end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2+7^2}$ or $\sqrt{16+49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** $\vec{AC}=\begin{pmatrix}7-3\\1-(-2)\end{pmatrix}$ or $\begin{pmatrix}4\\3\end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB}=\begin{pmatrix}-8\\y+2\end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2+(y+2)^2}=17$ **[M1]**

$64+(y+2)^2=289$ or $(y+2)^2=225$ **[M1]**

$y+2=15$ or $y+2=-15$ **[M1]**

$y=13,\ y=-17$ **[A1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** $\mathbf{u}+\mathbf{v}=\begin{pmatrix}3k+1\\k-1\end{pmatrix}$ **[M1]**

$(3k+1)^2+(k-1)^2=(\sqrt{50})^2$ **[M1]**

$9k^2+6k+1+k^2-2k+1=50$ **[M1]**

$10k^2+4k+2=50 \implies 5k^2+2k-24=0$ **[A1]**

(b) $(5k+a)(k+b)=0$ where $ab=-24$ or $5b+a=2$, or correct substitution into quadratic formula **[M1]**

$(5k+12)(k-2)=0$ **[M1]**

$k=2$ **[A1]**

$k=-2.4$ or $-\dfrac{12}{5}$ **[A1]**

(c) Substituting $k=2$ and $k=-2.4$ into $\begin{pmatrix}2k\\k-3\end{pmatrix}$ **[M1]**

$\begin{pmatrix}4\\-1\end{pmatrix}$ and $\begin{pmatrix}-4.8\\-5.4\end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

4 $\sqrt{5^2+(-12)^2}$ or $\sqrt{25+144}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 2]*

5 **(a)** $\begin{pmatrix} 9-2 \\ 4-(-3) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2+7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2+4^2}=5$ or $k^2+4^2=5^2$ **[M1]**

$k^2=25-16$ or $k^2=9$ **[M1]**

$k=3$, $k=-3$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $\sqrt{x^2+(x+2)^2}=10$ or $x^2+(x+2)^2=10^2$ **[M1]**

$x^2+x^2+4x+4=100$ **[M1]**

$2x^2+4x-96=0$ leading to $x^2+2x-48=0$ with no errors seen **[A1]**

(b) $(x+8)(x-6)=0$ or correct substitution into quadratic formula **[M1]**

$x=-8, x=6$ **[A1]**

(c) $\mathbf{u}=\begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u}=\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 7]*